IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

SEMICONDUCTOR DEVICES

Thursday, 7 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer question ONE.

Answer any TWO of the remaining questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : Z. Durrani
Second Marker(s) : K. Fobelets

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Constants

Permittivity of free space:	$\varepsilon_o = 8.85 \times 10^{-12}$ F/m
Permeability of free space:	$\mu_o = 4\pi \times 10^{-7}$ H/m
Intrinsic carrier concentration in Si:	$n_i = 1.45 \times 10^{10}$ cm^{-3} at $T = 300$ K
Dielectric constant of Si:	$\varepsilon_{Si} = 11$
Dielectric constant of SiO_2:	$\varepsilon_{ox} = 4$
Thermal voltage:	$kT/e = 0.026$ V at $T = 300$ K
Charge of an electron:	$e = 1.6 \times 10^{-19}$ C
Planck's constant:	$h = 6.63 \times 10^{-34}$ Js
Boltzmann constant:	$k = 1.38 \times 10^{-23}$ J/K
Band gap of Si:	$E_G = 1.12$ eV at $T = 300$ K
Electron affinity of Si:	$\chi = 4.05$ eV at $T = 300$ K
Effective density of states of Si:	$N_C = 3.2 \times 10^{-19}$ cm^{-3} at $T = 300$ K $N_V = 1.8 \times 10^{-19}$ cm^{-3} at $T = 300$ K

Formulae

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+V(x)\psi(x)=E\psi(x)$$ Schrödinger's equation in one dimension

$$f(E)=\frac{1}{1+\exp\left(\dfrac{E-E_f}{kT}\right)}$$ Fermi distribution

$$m_e^*=\frac{\hbar^2}{d^2E(k)/dk^2}$$ Effective mass

$$n=\frac{1}{\sqrt{2}\hbar^3}\left(\frac{m_e^*kT}{\pi}\right)^{3/2}e^{\left(-\frac{E_c-E_F}{kT}\right)}=N_ce^{\left(-\frac{E_c-E_F}{kT}\right)}$$ Concentration of electrons

$$p=\frac{1}{\sqrt{2}\hbar^3}\left(\frac{m_h^*kT}{\pi}\right)^{3/2}e^{\left(-\frac{E_F-E_v}{kT}\right)}=N_ve^{\left(-\frac{E_F-E_v}{kT}\right)}$$ Concentration of holes

$$\left.\begin{aligned}J_n(x)&=e\mu_n n(x)E(x)+eD_n\frac{dn(x)}{dx}\\J_p(x)&=e\mu_p p(x)E(x)-eD_p\frac{dp(x)}{dx}\end{aligned}\right\}$$ Drift and diffusion current densities in a semiconductor

$$I_{DS}=\frac{\mu C_{ox}W}{L}\left((V_{GS}-V_{th})V_{DS}-\frac{V_{DS}^2}{2}\right)$$ Current in a MOSFET

$$\left.\begin{aligned}J_n&=\frac{eD_n n_{p_0}}{L_n}\left(e^{\frac{eV}{kT}}-1\right)\\J_p&=\frac{eD_p p_{n_0}}{L_p}\left(e^{\frac{eV}{kT}}-1\right)\end{aligned}\right\}$$ Diffusion current densities in a pn-junction

$$V_0=\frac{kT}{e}\ln\left(\frac{N_AN_D}{n_i^2}\right)$$ Built-in voltage

$$c=c_0\exp\left(\frac{eV}{kT}\right)\text{ with }\begin{cases}c=p_n\text{ or }n_p\\c_0\text{ bulk minority carrier concentration}\end{cases}$$ Minority carrier injection under bias V

$$D=\frac{kT}{e}\mu$$ Einstein relation

$$w_n=\sqrt{\frac{2\varepsilon_0\varepsilon_r}{q}\frac{N_A}{N_AN_D+N_D^2}(V_0-V)}$$

$$w_p=\sqrt{\frac{2\varepsilon_0\varepsilon_r}{q}\frac{N_D}{N_AN_D+N_A^2}(V_0-V)}$$ Depletion width in the n and p-type region, under bias V

$$\delta c = \Delta c \exp\left(\frac{-x}{L}\right) \text{ with } \begin{cases} \delta c = \delta p_n \text{ or } \delta n_p \\ \Delta c \text{ the excess carrier concentration} \\ \text{at the edge of the depletion region} \end{cases}$$

Excess carrier concentration as a function of distance when recombination occurs – long layer approximation.

$$L = \sqrt{D\tau}$$

Diffusion length

$$C_{diff} = \frac{e}{kT} I\tau$$

Diffusion capacitance

$$i(t) = \frac{Q(t)}{\tau} + \frac{dQ(t)}{dt}$$

Time variation of current and charge

$$\frac{dE}{dx} = \frac{\rho(x)}{\varepsilon}$$

Poisson equation in 1 dimension

$$n = N_C \, exp\left(\frac{E_F - E_C}{kT}\right)$$

$$p = N_V \, exp\left(\frac{E_V - E_F}{kT}\right)$$

Carrier concentrations

1. [This question carries 40% marks]

a)

i) Give the relation between the intrinsic hole concentration and the intrinsic electron concentration.

[2]

ii) Give the hole and electron concentration in Boron doped Si (*p*-Si) with a doping concentration of 2×10^{23} m^{-3}.

[2]

iii) The measured conductivity of the *p*-Si material in (ii) is 15.5 S/cm at 300 K. Calculate the mobility of the majority carriers in this material.

[4]

iv) Demonstrate why in the material of (ii), the flow of electrons can be ignored.

[4]

b) At low electric fields, the drift current through an *n*-type semiconductor is given by the expression: $I = en\mu EA$, where e is the unit charge, n the electron concentration, μ the mobility, E the electric field and A the cross-sectional area. Which parameters in this equation are influenced when the donor doping density in the semiconductor is increased by a factor of 100? Give the reason for your answer.

[4]

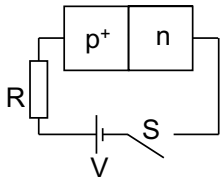


Figure 1.1

c) A p^+n diode, in series with a load resistance R and voltage V, is switched from an open circuit condition (Figure 1.1) by switch S to a closed-circuit condition.

i) By considering the time variant relationship between current and charge, show that the time dependence of hole charge Q_p building up in the diode is given by:

$$Q_p(t) = \tau_p I_{ON}\left(1 - \exp\left(-\frac{t}{\tau_p}\right)\right)$$

where τ_p is the hole minority carrier lifetime.

[6]

ii) Switch S is now opened to switch the diode from ‘on’ to open circuit ‘off’. Show that the time dependence of Q_p is now given by:

$$Q_p(t) = I_{ON}\tau_p \exp\left(\frac{-t}{\tau_p}\right)$$

[6]

d) Sketch the band diagram (E_c, E_v, E_F, E_G), from the emitter to the collector, for a n^+pn bipolar junction transistor (BJT) with:

i) The BJT at equilibrium.

[4]

ii) The BJT in saturation.

[4]

iii) The BJT in reverse active mode.

[4]

2. [This question carries 30% marks]

a) A *pn* diode is reverse biased at voltage $V < V_{BD}$, where V_{BD} is the breakdown voltage. Assume impact ionisation occurs only due to electrons. What is the ratio between the density of electrons injected into the depletion region to the density of electron leaving the depletion region, if the depletion width is 20 times the electron scattering length?

[6]

b) Consider a cylindrical *pn* diode with cross-sectional diameter d = 100 μm. The doping densities in the *p* and *n* regions are respectively $N_A = 2 \times 10^{18}$ cm^{-3} and $N_D = 10^{16}$ cm^{-3}. Calculate, at T = 300 K, values for following quantities.

i) Built-in voltage V_0 .

[4]

ii) Depletion widths w_n and w_p in the *n* and *p* regions.

[6]

iii) Charge Q in the depletion region, on each side of the junction.

[6]

iv) The minimum electric field.

[8]

3. [This question carries 30% marks]

a) The cross section for a short p^+np BJT, in forward active mode, is shown in Figure 3.1. White regions are neutral regions, grey regions are depletion regions, and dashed lines indicate positions of metallurgical junctions. The metallurgical junction between the base (B) and collector (C) lies at $x = X_B$ and that between the base and emitter (E) at $x = 0$. The depletion region widths in the base and the collector at the BC junction are w_{Br} and w_{Cr} respectively. The corresponding depletion region widths in the emitter and base at the EB junction are w_E and w_{Bf} respectively.

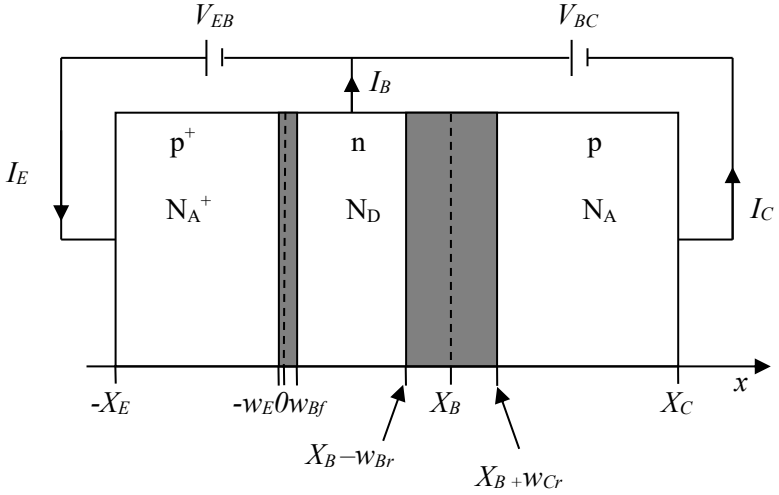


Figure 3.1

i) Give a simple relationship between the emitter current I_E, base current I_B and collector current I_C.

[4]

ii) Derive expressions for I_E, I_B and I_C as a function of material parameters, taking into account the effect of the depletion region. State any assumptions you make.

[10]

b) Give the definition for transconductance g_m of the p^+np BJT and hence, derive an expression for g_m as a function of the material parameters.

[6]

c) By considering excess minority carrier injection across the EB junction, show that the EB junction capacitance C is given by:

$$C \approx \frac{L_p^2 I_{EB}}{2 D_p V_t}$$

where L_p and D_p are the minority carrier diffusion length, and diffusion constant for holes, respectively, I_{EB} is the EB current, and V_t is the thermal voltage.

[10]

4. [This question carries 30% marks]

The cross-section through a *p*-channel, enhancement mode Si MOSFET is shown in Figure 4.1. The device has gate length L_g, gate width W_g, gate oxide thickness t_{ox} and gate capacitance per unit area C_{ox}. The drain-source and gate-source voltages are V_{ds} and V_{gs} respectively, the MOSFET threshold voltage is V_{th}, and the source is grounded.

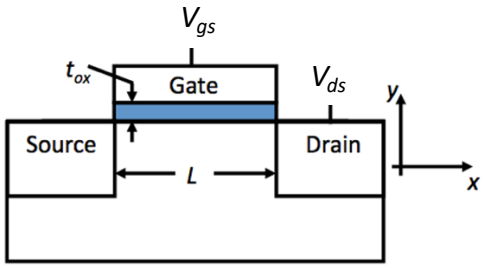


Figure 4.1

a) (i) What is the doping type in the source, drain regions and substrate regions?

[4]

(ii) Is the threshold voltage V_{th} greater than or less than zero?

[2]

(iii) For the MOSFET in depletion, in what range does V_{gs} lie? Choose from the following four options.

$0 < V_{gs} < V_{th}$

$0 < V_{th} < V_{gs}$

$0 > V_{th} > V_{gs}$

$0 > V_{gs} > V_{th}$

[2]

b) Sketch the energy band diagram (E_c, E_v, E_F, E_G) through the channel of the MOSFET of part (a), between source and drain, for:

i) The MOSFET in equilibrium.

ii) The MOSFET biased in the triode region.

iii) The MOSFET biased in the saturation region, beyond pinch-off.

[2+2+4]

c) The current-voltage characteristics of a p-channel MOSFET are given by the following equation:

$$I_{ds} = \frac{\mu_p C_{ox} W_g}{L_g}\left[\left(|V_{gs}| - |V_{th}|\right)|V_{ds}| - \frac{V_{ds}^2}{2}\right]$$

Derive expressions for transconductance g_m in the triode and saturation regions.

[4 + 4]

d) Draw a cross-sectional diagram for a double-diffused MOS transistor (DMOS), along a plane cutting through the source, drain and gate regions. Your diagram should show the device layer structure, and the position of the source, drain, gate and gate oxide regions.

[6]

[END]